

**THE EBR OPERATOR AT $d = 2$ IS NOT A G -OPERATOR
AND ITS DIFFERENTIAL GALOIS GROUP IS $\mathrm{SL}(4)$:
ARITHMETIC, IRREDUCIBILITY, AND THE
NON-LIOUVILLIAN COROLLARY**

PAPANOKECHI

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Independent Researcher (ORCID 0009-0000-6192-8273). This is a preprint; nothing herein is a claim of priority. Every load-bearing statement is graded PROVEN/STRUCTURAL/VERIFIED/CONJECTURED (§1.2); PROVEN is reserved for the Lean 4 cores of §9.

ABSTRACT. For the positive-coefficient polynomial continued fraction with $b(n) = 3n^2 + n + 1$, the Edge–Borel object $G(s) = \sum_{n \geq 0} Q_n s^n / (2n)!$ is annihilated by an explicit order-4 linear differential operator L_2 over $\mathbb{C}(s)$ with regular singular points at $s = 0$ and $s = R = 4/3$ and an *irregular* singularity at $s = \infty$ of slope $1/4$. We establish four things about L_2 at increasing evidential strength. First, in the Borel normalization $g_n = Q_n / (2n)!$ the operator is *not* the minimal operator of a G -function: the Taylor denominators retain essentially the full factorial, the p -curvature is non-nilpotent for every tested prime, and the irregular point at infinity is incompatible with the André–Chudnovsky–Katz trichotomy. Second, L_2 is irreducible and minimal, by two independent arguments. Third, its local monodromy at R is a *semisimple* complex pseudo-reflection with spectrum $\{1, 1, 1, e^{i\pi/3}\}$ — correcting an earlier “resonance logarithm” reading, consistent with the criterion that a $-\gamma$ logarithm requires $\gamma \in \mathbb{Z}$ whereas here $\gamma = 11/6$. Fourth, and centrally, the identity component of the differential Galois group is $G_{\text{Gal}}(L_2)^\circ = \text{SL}_4$; we obtain this by eliminating every class of the maximal-subgroup (Aschbacher) classification of algebraic subgroups of SL_4 , with the imprimitive class closed unconditionally through a unique quadratic-character analysis and an *a priori* degree bound that makes the underlying rational-solution searches complete. The immediate corollary is that L_2 has no Liouvillian solutions. The connection coefficient is computed by a third independent channel to 169 stable digits; a positive-control-validated integer-relation battery finds, to that precision, that it is neither an elementary Γ -quotient nor a low-height combination of standard constants nor algebraic of small degree. We state with care what these results do *not* give: a large Galois group does not imply the connection coefficient is transcendental, connection coefficients are not differential-Galois invariants, and the transcendence question therefore remains a conjecture whose only route is a separate period analysis. The degree-bound arithmetic, the pullback exponent arithmetic, the eigenvalue-parity bookkeeping, and the parity of the relevant 4-cycle are formalized in Lean 4/Mathlib with audited axiom cones.

1. INTRODUCTION

1.1. Background and the object of study. This paper continues the Edge–Borel Radius (EBR) program. EBR-I established that for a positive-coefficient polynomial continued fraction $Q_n = b(n)Q_{n-1} + Q_{n-2}$ with $b(n) = \beta_d n^d + \dots$, the physical Borel transform $G(s) = \sum_{n \geq 0} Q_n s^n / (dn)!$ is holonomic with a dominant singularity at $s = R = d^d / \beta_d$ and local form $G(s) \sim A (1 - s/R)^{-\gamma}$, with $\gamma = (d+1)/2 + b_{d-1}/\beta_d$ [1]. EBR-II identified the amplitude A as a *connection coefficient* of the order- $2d$ operator annihilating G — the entry of the connection matrix linking the exponent-0 local solution at the origin to the dominant exponent- $(-\gamma)$ solution at $s = R$ — and conjectured that this coefficient is generically transcendental, with a “rigidity dividing line” separating the rigid case $d = 1$ from the non-rigid cases $d \geq 2$ [2].

We fix the smallest non-rigid case. Throughout, $d = 2$ and

$$b(n) = 3n^2 + n + 1, \quad \beta_2 = 3, \quad g_n = \frac{Q_n}{(2n)!}, \quad G(s) = \sum_{n \geq 0} g_n s^n, \quad (1)$$

so that $R = 4/3$ and $\gamma = 11/6$. The annihilating operator, written with $\theta = s d/ds$ as $L = P_0(\theta) - s P_1(\theta + 1) - s^2$ and cleared to the form $\sum_{k=0}^4 a_k(s) d^k/ds^k$, has

$$\begin{aligned} a_4 &= 4s^4(4 - 3s), & a_3 &= 48s^3 - 94s^4, & a_2 &= 12s^2 - 156s^3, \\ a_1 &= -30s^2, & a_0 &= -s^2. \end{aligned} \quad (2)$$

We call this operator L_2 . Its leading coefficient $a_4 = d^d s^{2d}(d^d - \beta_d s)$ matches EBR-I, and its finite singular locus is exactly $\{0, R\}$ [CC1-OP-L2].

1.2. Results and their evidential status. We organize the paper by a four-class evidence discipline, stated here so that the reader can audit each assertion against the kind of evidence that supports it. `PROVEN` is reserved for results carrying a machine-checked Lean 4 proof with an audited axiom cone; `STRUCTURAL` denotes a complete mathematical argument verified by hand and computer algebra but not formalized; `VERIFIED` denotes a reproducible high-precision or exact symbolic computation; `CONJECTURED` denotes an open statement. A grade table appears in §10. The principal results are:

- (R1) *Arithmetic nature.* In the normalization (1), L_2 is not the minimal operator of a G -function (§3; VERIFIED/STRUCTURAL).
- (R2) *Irreducibility.* L_2 is irreducible and is the minimal annihilator of G (§4; STRUCTURAL, two routes).
- (R3) *Local monodromy.* The monodromy at R is a semisimple pseudo-reflection with spectrum $\{1, 1, 1, e^{i\pi/3}\}$ (§5; VERIFIED), correcting an earlier reading (§5.2).
- (R4) *Main theorem.* $G_{\mathrm{Gal}}(L_2)^\circ = \mathrm{SL}_4$ (§6; STRUCTURAL, bound-complete).
- (R5) *Corollary.* L_2 has no Liouvillian solutions (§7; VERIFIED-by-citation applied to (R4)).
- (R6) *Connection data and a null.* The connection coefficient is computed to 169 digits and is, to that precision, not an elementary Γ -quotient, not a low-height constant combination, and not algebraic of small degree (§8; VERIFIED).

1.3. What we do not claim. We are explicit about the limits of the method, since the temptation to overread a large Galois group is precisely the error the EBR program guards against.

Non-rigidity ($P = d - 1 > 0$) does not imply that the connection coefficient is transcendental; a large differential Galois group does not imply that the connection coefficient is transcendental. The present work targets the group only. Connection coefficients are not differential-Galois invariants,

and the transcendence of the connection coefficient remains a conjecture whose only route is a separate period analysis.

Accordingly the abstract and this introduction state the SL_4 theorem and the not-a- G -function result, and no transcendence claim of any kind. The G -function statement is always qualified by its normalization scope; the group statement is for $d = 2$, with general d flagged open (§11); the explicit numerical Stokes multipliers at the ramified irregular point are out of scope, for the reason given in §8.2.

2. THE OPERATOR AND ITS SINGULAR STRUCTURE

2.1. Riemann data at the regular points. At $s = 0$ and $s = R = 4/3$ the operator L_2 is regular singular. The indicial computation gives

$$\begin{aligned} \exp_0 &= \left\{ \frac{j}{d} : j = 0, \dots, 2d-1 \right\} = \left\{ 0, \frac{1}{2}, 1, \frac{3}{2} \right\}, \\ \exp_R &= \{0, 1, 2, -\gamma\} = \left\{ 0, 1, 2, -\frac{11}{6} \right\}, \end{aligned} \quad (3)$$

with $\gamma = (d+1)/2 + b_{d-1}/\beta_d = 11/6$ [CC1-RIEMANN]. The dominant exponent $-\gamma$ differs from each of the integers $\{0, 1, 2\}$ by a non-integer, a fact that will decide the local monodromy type in §5.

2.2. The irregular point at infinity. The point at infinity is irregular. The Fuchs degree bound $\deg a_k \leq \deg a_4 - (4 - k)$ holds for $k = 1, \dots, 4$ but *fails* at $k = 0$, where $\deg a_0 = 2 > 1$; the violation is caused exactly by the term $a_0 = -s^2$. The Newton polygon at infinity has a single edge from $(0, 2)$ to $(4, 5)$ of slope $3/4$, with exact edge polynomial $-12c^4 - 1$; its roots $c^4 = -1/12$ have $|c| = (1/12)^{1/4} = 0.53728\dots$ and arguments an odd multiple of $\pi/4$. There is a single Puiseux cycle of length 4, so the ramification is exactly 4, the slope is $1/4$, the integer Poincaré rank is 1, and the four determining factors

$$E_k(s) = \exp(4c\zeta_4^k s^{1/4}), \quad c^4 = -\frac{1}{12}, \quad \zeta_4 = i, \quad k = 0, 1, 2, 3, \quad (4)$$

form one transitive orbit under the cyclic order-4 ramification group. The prediction is corroborated numerically: at $s = 10^{14}$ the four scaled symbol-roots $w s^{3/4}$ have modulus tending to 0.53728 and arguments tending to $\{\pm\pi/4, \pm3\pi/4\}$ [CC1-INF-IRREGULAR]. This corrects the earlier “three regular singular points” reading of the operator.

2.3. Accessory count and non-rigidity. Because infinity is irregular, the Fuchsian accessory count $N_{\mathrm{acc}} = (2d-1)(d-1)$ of EBR-II is inapplicable; the corrected index-of-rigidity count is $P = d-1$, which at $d = 2$ equals 1. We do not re-derive the full Katz rigidity index with the irregular point here; this is flagged as a residual gap (§11). The point that matters downstream is robust to the discrepancy: the count is positive at $d = 2$ on either reading, so the connection problem is non-rigid [CC1-ACCESSORY].

3. ARITHMETIC NATURE: NOT A G -OPERATOR

We emphasize at the outset the *normalization scope* of this section: all statements are about the Borel normalization $g_n = Q_n/(2n)!$ of (1) and do not extend to arbitrary rescalings of the coefficient sequence (multiplying by $(2n)!$, for instance, returns the integer series $\sum Q_n s^n$ of radius 0, a different object). With that scope fixed:

3.1. Integrality and denominator growth. The numerators are integers: $Q_0 = 1$, $Q_1 = 5$, and $Q_n = (3n^2 + n + 1)Q_{n-1} + Q_{n-2}$ has integer coefficients, so $Q_n \in \mathbb{Z}$ (checked to $n \leq 2000$), whence $\mathrm{denom}(g_n) \mid (2n)!$ and $q_N := \mathrm{lcm}_{n \leq N} \mathrm{denom}(g_n) \mid (2N)!$ [CC2-0-QINT]. The content of the section is that this divisibility is *not* far from an equality: q_N grows superlinearly, the fit $q_N \sim N \log N$ giving $R^2 = 0.9999983$ against $R^2 = 0.998858$ for a linear law, and $\log_{10} q_N / \log_{10}(2N)! \approx 1.0000$. The Taylor denominators retain essentially the entire factorial; there is no geometric denominator collapse, so G is not a G -function in the André–Bombieri sense, but a Gevrey/Borel object [CC2-0-GFUNC].

3.2. Two confirming channels. The p -curvature ψ_p of L_2 over $\mathbb{F}_p(s)$ is non-nilpotent for every tested prime $p \in \{5, 7, 11, 13, 17, 19, 23, 29\}$, at three generic specializations each (the primes 2, 3 are excluded as degenerate for the leading coefficient). Since global nilpotence is equivalent to nilpotence of ψ_p for almost all p , the operator is not globally nilpotent, which by the Chudnovsky–André theorem again excludes the G -function property and matches the irregular-infinity prediction of Katz [CC2-0-PCURV]. Indeed the André–Chudnovsky–Katz results yield a clean trichotomy: of the three statements “ G is a G -function”, “ L_2 is minimal/irreducible”, and “ ∞ is irregular”, at most two can hold. We establish the latter two (§4 and §2.2), so the first fails [CC2-0-TRICHOTOMY].

3.3. Consequence for the period route. A direct corollary concerns the transcendence question. Because G itself is not a G -function, no unconditional transcendence statement about the connection coefficient can be drawn from “ G is a G -function with the right growth”. The G -function/period machinery may still bear on the coefficient as a connection datum of the order-4 operator that is Fuchsian at $\{0, R\}$ and irregular at infinity, but establishing that is the burden of the separate period analysis and cannot be inherited as assumed [CC2-0-CC3-AMEND].

4. IRREDUCIBILITY AND MINIMALITY

The falsification target “ L_2 is reducible over $\mathbb{C}(s)$ ” is refuted by two independent arguments.

4.1. Formal module at infinity. At the irregular point the formal $\mathbb{C}((s^{-1}))$ -module of L_2 has a single slope $1/4$ whose four determining factors (4) form one transitive orbit under the cyclic order-4 ramification group. A proper right factor over $\mathbb{C}(s)$ would have a $\mathbb{C}(s)$ -rational, hence ramification-stable, set of determining factors; but the only subsets of a transitive size-4 orbit stable under the generating 4-cycle are the empty set and the whole set. Hence L_2 has no proper right factor and is the minimal annihilator of G (order $4 = 2d$). The local-to-global inference uses Turrittin’s formal classification and the formal-module description of [3, Ch. 3], and is graded STRUCTURAL (computer algebra and hand argument; an independent DFactor-style confirmation was unavailable on the host) [CC1-IRREDUCIBLE].

4.2. Exhaustive low-order factor exclusion. Independently, slope additivity at infinity (the Newton polygon of a product is the concatenation of the factors’ polygons) reduces the existence of an order- m right factor to the existence of a σ -stable size- m subset of the determining-factor orbit, where $\sigma = (0\ 1\ 2\ 3)$ generates the ramification group. A finite enumeration shows the stable subset sizes are $\{0, 4\}$, so no order-1, 2, or 3 right factor exists; for order 1, each factor in (4) is ramified, so there is no single-valued $\exp \int r$ with $r \in \mathbb{C}(s)$. This recovers irreducibility by an independent route [CC2-0-FACTOR]. The two routes are logically independent — one local-analytic at infinity, one a finite combinatorial enumeration of σ -stable orbits — and neither relies on a decision-procedure factorization; we grade irreducibility STRUCTURAL on the strength of their agreement, and it is this irreducibility that the degree-bound completeness of §6.5 later makes load-bearing.

5. LOCAL MONODROMY AND THE SEMISIMPLE CORRECTION

5.1. The monodromy at R is a semisimple pseudo-reflection. The local monodromy at R has eigenvalues $e^{2\pi i \rho}$ for $\rho \in \exp_R = \{0, 1, 2, -\frac{11}{6}\}$, that is $\{1, 1, 1, e^{i\pi/3}\}$ (using $e^{-2\pi i \cdot 11/6} = e^{i\pi/3}$). Although the eigenvalue 1 occurs with multiplicity 3 and there is an integer-exponent resonance among $\{0, 1, 2\}$, no logarithm forms: an exact symbolic Frobenius solve at R shows that all resonance obstructions on the eigenvalue-1 tower vanish, giving three independent logarithm-free solutions, while an independent numerical computation gives $\text{rank}(M_R - I) = 1$ (one singular value 2.159, the rest below 10^{-30}). Thus M_R is a semisimple complex pseudo-reflection of order 6 with spectrum $\{1, 1, 1, e^{i\pi/3}\}$, and M_0 is semisimple with spectrum $\{1, -1, 1, -1\}$ [CC2-2D-JORDAN][CC4-1-MONODROMY]. The exact-arithmetic resonance residuals were subsequently pushed to 169 digits, confirming the absence of a logarithm to that precision [CC4-1-NOLOG].

5.2. Erratum. This corrects an earlier narration that spoke of a “unipotent part” or “resonance logarithm” at R . The corrected statement is that M_R is a semisimple pseudo-reflection with no logarithm, consistent with the criterion of [2] that a $-\gamma$ resonance logarithm requires $\gamma \in \mathbb{Z}$;

here $\gamma = 11/6 \notin \mathbb{Z}$. The superseded reading appeared in the narration of EBR-II (concept DOI 10.5281/zenodo.20566465, current version 10.5281/zenodo.20571232, v1.2); the load-bearing claims of that work — the exponents, the irreducibility, and the irregularity at infinity — are unaffected, only the adjective on M_R [CC4-ERR-1].

6. THE DIFFERENTIAL GALOIS GROUP

We compute $G_{\mathrm{Gal}}(L_2)$ over $\mathbb{C}(s)$. The strategy is the maximal-subgroup classification of algebraic subgroups of SL_4 in the differential-Galois setting, following the formulation of [3, Ch. 4]; we use the Aschbacher class labels $C_1, \dots, C_8, S$ as organizing vocabulary for the maximal closed subgroups of SL_4 , and we stress that this is the algebraic-group classification, not an application of Aschbacher's finite-group theorem. In the geometric (Kleidman–Liebeck) numbering the classical-form subgroups $\mathrm{SO}_4, \mathrm{Sp}_4$ are class C_8 , the subfield class is C_5 , the extraspecial-normalizer class is C_6 , and the tensor-induced class is C_7 . Over the algebraically closed field of constants in characteristic zero the subfield class C_5 is empty; the tensor-induced class C_7 for the $2 \otimes 2$ decomposition shares its identity component $\mathrm{SL}_2 \otimes \mathrm{SL}_2$ with C_4 and is excluded with it; and the extraspecial-type class C_6 is finite, hence killed together with the other finite primitive classes by the exponential torus. The operative algebraic classes are therefore $C_1, C_2, C_3, C_4, C_8, S$ together with the finite classes.

6.1. Determinant and the SL_4 normalization. The Wronskian satisfies $w'/w = -a_3/a_4 = (24 - 47s)/(2s(3s - 4)) = -3/s + (29/2)/(4 - 3s)$, so $W = s^{-3}(4 - 3s)^{-29/6}$, with determinant exponents -3 at 0 and $-29/6$ at R (each cross-checked by $W_{\mathrm{exp}} = \sum \exp -\binom{4}{2}$ with residual 0). The SL_4 -normalizing twist $\chi = \det^{1/4} = s^{-3/4}(4 - 3s)^{-29/24}$ is algebraic over $\mathbb{C}(s)$ (a product of rational powers; not rational, not properly Liouvillian). After twisting by χ the regular-point exponent multisets become $\{3/4, 5/4, 7/4, 9/4\}$ at 0 and $\{29/24, 53/24, 77/24, -5/8\}$ at R , each summing to 6 [CC2-1-TWIST]. The four determining-factor leading units $c \cdot \{1, i, -1, -i\}$ at infinity generate a rank-2 lattice $\mathbb{Z} \cdot 1 + \mathbb{Z} \cdot i$, so the formal exponential torus has dimension 2 ; in particular $G_{\mathrm{Gal}}(L_2)^\circ$ contains a 2-dimensional torus and is infinite [CC2-1-EXPTORUS].

6.2. No invariant form: C_8 ($\mathrm{SO}_4, \mathrm{Sp}_4$). A nondegenerate G -invariant bilinear form, even up to a rank-1 twist $V \cong V^* \otimes \kappa$, would force the local eigenvalue multiset at R to equal $\kappa \cdot \{\text{inverses}\}$. The multiplicity-3 eigenvalue 1 pins $\kappa = 1$, and then $e^{i\pi/3} \neq e^{-i\pi/3}$. Hence the multiset $\{1, 1, 1, e^{i\pi/3}\}$ is not inversion-closed up to any scalar, so L_2 carries no invariant symmetric or alternating form, even up to twist: the group lies in neither SO_4 nor Sp_4 . This argument is exact and uses only the $-11/6$ exponent [CC2-2A-SELF DUAL]. It is confirmed constructively: the exterior square $\Lambda^2 L_2$ has order 6 (no symplectic order-drop), splitting at infinity into a rank-2 slope-0

piece — the cancelling pairs $(0, 2)$ and $(1, 3)$, since $i^0 + i^2 = 0$ and $i^1 + i^3 = 0$ — and a rank-4 slope-1/4 remainder; the search for a rational antisymmetric solution of $P' = -A^\top P - PA$ returns nullspace dimension 0 in two ansatz boxes (denominator $s^a(4 - 3s)^b$, numerator degree d ; $(a, b, d) = (4, 4, 10)$ and $(3, 3, 8)$), so the group is not in Sp_4 [CC2-2B-SP4]. The same search for a symmetric solution returns nullspace dimension 0 in the same boxes, so the group is not in SO_4 [CC2-2C-SO4].

6.3. Tensor and tensor-induced classes: C_4, C_7, S . A symmetric cube $\mathrm{Sym}^3 W$ of a rank-2 module is self-dual up to twist, contradicting §6.2; moreover a rank-2 module has ramification at most 2 and cannot carry slope 1/4, and Sym^3 would produce factors $\{3\phi, \phi, -\phi, -3\phi\}$ of two distinct magnitudes rather than the single equal-magnitude transitive $\mathbb{Z}/4$ orbit of L_2 . Hence the $\mathrm{Sym}^3 \mathrm{SL}_2$ class S is excluded. A tensor product $V_1 \otimes V_2$ of two rank-2 modules (class C_4) is likewise self-dual up to twist and again cannot produce slope 1/4, so it too is excluded; the tensor-induced class C_7 (the wreath $\mathrm{SL}_2 \wr S_2$ acting on the $2 \otimes 2$ decomposition) has the same identity component $\mathrm{SL}_2 \otimes \mathrm{SL}_2$ and is excluded by the same argument [CC2-2D-TENSOR].

6.4. The imprimitive classes C_2, C_3 and primitivity. The remaining obstruction is imprimitivity: the monomial/2-block class C_2 and the field-extension class C_3 . These are not separable from the local data alone, because the connected exponential torus fixes every block of any system of imprimitivity, forcing blocks to be spans of weight lines, and both the 4-cycle formal monodromy and the semisimple pseudo-reflection M_R preserve the 2-block system $\{\{L_1, L_3\}, \{L_2, L_4\}\}$ [CC2-2D-IMPRIM]. We close the class globally.

A quadratic character η realizing an imprimitive structure ramifies at an even number of singular points. Since the R -spectrum $\{1, 1, 1, e^{i\pi/3}\}$ is not negation-closed (no -1), η must be unramified at R ; evenness then forces ramification exactly at $\{0, \infty\}$, i.e. $\eta = \sqrt{s}$, which passes the necessary local tests because the spectra at 0 ($\{1, -1, 1, -1\}$) and at ∞ (units $\{1, i, -1, -i\}$) are negation-closed. Thus $\sqrt{s}$ is the unique candidate character [CC4-A1-ETA-UNIQUE]. Furthermore the formal monodromy at infinity is the 4-cycle $(q_0 q_1 q_2 q_3)$ on the determining factors (since $s^{1/4} \mapsto i s^{1/4}$), an odd permutation; any induced structure embeds the permutation action in a transitive subgroup of S_4 containing this 4-cycle, hence in $\{C_4, D_4, S_4\}$, each of which has an index-2 subgroup, i.e. a quadratic character, which by uniqueness is $\sqrt{s}$. This reduces the 2-block, the 4-line monomial, and the degree-2/4 field-extension classes all to the single $\sqrt{s}$ test [CC4-A1B-MONOMIAL-CLOSURE].

That test is negative, by two independent searches. Over $\mathbb{C}(s)$, the search for a rational intertwiner Φ realizing $M \cong M \otimes \sqrt{s}$ (solving $\Phi' = A\Phi - \Phi A + (1/2s)\Phi$) returns nullspace dimension 0 across boxes $s^a(4 - 3s)^b$ of degree up to 20 (336 unknowns at the largest); as a calibration, the untwisted search returns $\mathrm{End}_{\mathbb{C}(s)}(M) = 1$ (scalars only), confirming absolute irreducibility and that the pipeline detects solutions when they exist [CC4-0-ROUTE2-PRIMITIVE].

Independently, the pullback under $s = t^2$ gives $\tilde{A}(t) = 2t A(t^2)$, with $t = 0$ exponents doubling to $\{0, 1, 2, 3\}$, slope $1/2$ at infinity, and two unramified points $\pm t_R$ swapped by the deck map; the eigenring of the pullback is 1-dimensional (scalars), so the pullback is irreducible over $\mathbb{C}(t)$ and M is not induced from the quadratic cover [CC4-0-ROUTE1-PRIMITIVE]. The two routes agree on primitivity.

6.5. Completeness of the searches. The two rational-solution searches are made complete by an *a priori* degree bound. Because M_0 and M_R are semisimple (no logarithm), a rational horizontal section is single-valued and every non-integer End-exponent branch is killed, so the pole order at a finite singular point p is at most $-\min$ of the integer End-exponents at p (these exponents being pairwise differences of the M -exponents, shifted by the η -twist), and the growth at infinity is bounded by the minimal integer slope-0 exponent of the ramified formal monodromy. Computed exactly, the numerator-degree bound for Route 2 is $B_2 = 1 + 2 + 0 = 3$ (pole ≤ 1 at 0, ≤ 2 at R , no growth at infinity), and for Route 1 it is $B_1 = 3 + 2 + 2 = 7$ (pole ≤ 3 at $t = 0$, ≤ 2 at each $\pm t_R$). Both are at most 20, and the boxes actually run — $[s(4 - 3s)^2, \deg \leq 8]$ and $[t^3(3t^2 - 4)^2, \deg \leq 10]$ — strictly contain B_2 and B_1 . Hence the nulls of §6.4 are bound-complete, not merely empirical [CC4-0B-BOUNDS].

6.6. The classification and the main theorem. We collect the eliminations.

Class	Description	Eliminated by	Grade
C_1	reducible	§4 ([CC1-IRREDUCIBLE], [CC2-0-FACTOR])	S
$C_6/\text{fin.}$	extraspecial-type / finite	§6 torus ([CC2-1-EXPTORUS])	S
C_8	orthogonal SO_4	§6.2 ([CC2-2C-SO4], [CC2-2A-SELFDUAL])	V/S
C_8	symplectic Sp_4	§6.2 ([CC2-2B-SP4], [CC2-2A-SELFDUAL])	V/S
S	$\mathrm{Sym}^3 \mathrm{SL}_2$	tensor/slope ([CC2-2D-TENSOR])	S
C_4, C_7	$\mathrm{SL}_2 \otimes \mathrm{SL}_2$	tensor/slope ([CC2-2D-TENSOR])	S
C_2	monomial / 2-block	$\sqrt{s}$ test ([CC4-0-ROUTE2-PRIMITIVE], [CC4-0-ROUTE1-PRIMITIVE])	S
C_3	field extension	4-cycle closure ([CC4-A1B-MONOMIAL-CLOSURE])	S
C_5	subfield	vacuous over $\overline{\mathbb{C}}$, char 0	S

Theorem 1. *For the operator L_2 of (2), the identity component of the differential Galois group over $\mathbb{C}(s)$ is*

$$G_{\mathrm{Gal}}(L_2)^\circ = \mathrm{SL}_4.$$

Evidential status of the proof. The two imprimitivity-excluding routes ($\mathrm{Hom}(M, M \otimes \sqrt{s}) = 0$ over $\mathbb{C}(s)$; eigenring = scalars on the $s = t^2$ cover)

agree and are bound-complete by §6.5; the unique-character analysis ([CC4-A1-ETA-UNIQUE]) and the 4-cycle closure ([CC4-A1B-MONOMIAL-CLOSURE]) reduce every imprimitive class to the single $\sqrt{s}$ test; the remaining maximal classes are excluded as tabulated. With $C_1, \dots, C_8, S$ exhausted and the torus forcing the group to be infinite and reductive, the identity component is SL_4 . The graded inputs are STRUCTURAL (the routes, the character analysis, the bound) and VERIFIED-by-citation (the classification framework of [3]); the theorem is therefore STRUCTURAL and, with the bound of §6.5, unconditional [CC4-0-SL4-VERDICT]. $\square$

7. THE NON-LIOUVILLIAN COROLLARY AND THE BRIDGE

Corollary 2. *L_2 has no Liouvillian solutions.*

Proof. A linear operator has a Liouvillian solution if and only if the identity component of its differential Galois group is solvable [3, Ch. 1,4]. By Theorem 1 the identity component is SL_4 , which is connected, semisimple, and non-solvable; hence no Liouvillian solution exists. This is a citation applied to (R4) and is graded VERIFIED-by-citation + STRUCTURAL application [CC4-2-BRIDGE]. $\square$

We restate the discipline of §1.3 as the bridge to the open problem. The group computation licenses exactly the non-Liouvillian corollary and nothing about the arithmetic nature of the connection coefficient. Connection coefficients are not differential-Galois invariants; the transcendence of the connection coefficient remains a conjecture, and its only route is the separate period analysis, which must be argued afresh because the G -function clause is retired by §3.

8. CONNECTION DATA AND THE INTEGER-RELATION NULL

8.1. The connection coefficient to 169 digits. We computed the connection coefficient by a third independent channel — a full 4×4 local fundamental system at R in the basis $\{u_{-11/6}, u_0, u_1, u_2\}$, matched to the continued series G and its first three derivatives at an interior point — obtaining the amplitude

$$A = 2.87436850995728075034795582039806005699192841843121 \dots$$

and the coefficient-asymptotics prefactor

$$C_{\mathrm{EBR}} = \frac{A}{\Gamma(11/6)} = 3.055706807890481365701912201727 \dots,$$

the latter appearing in $g_n \sim C_{\mathrm{EBR}} R^{-n} n^{\gamma-1}$. The channel is self-validated by multi-point agreement ($s^* = 1.0$ versus 0.9, agreeing to 169 digits) and multi-precision agreement (working precision 170 versus 210, agreeing to 169 digits); a Richardson channel on the coefficient asymptotics independently confirms 77 digits, the clean $1/n$ extrapolation being itself a re-confirmation of the absence of a logarithmic tower at R . We state honestly that these are

`mpmath` floating values, not formal interval enclosures; the grade is VERIFIED, not PROVEN [CC4-1-C-120D].

8.2. Stokes structure, and what is out of scope. At the irregular point the four determining factors $q_k = \gamma_k s^{1/4}$ with $\gamma_k^4 = -1/12$ give eight singular directions in the $w = s^{1/4}$ plane, spaced $\pi/4$, and a formal monodromy equal to a single 4-cycle (permutation sign -1) times $\text{diag}(e^{2\pi i \mu_k})$. Determinant consistency of the cyclic relation $M_0 M_R M_\infty = I$ forces $\det M_\infty = e^{-i\pi/3}$, i.e. $\sum_k \mu_k \equiv 1/3 \pmod{1}$, a constraint any Stokes computation must satisfy. We are explicit that the *numerical* Stokes multipliers at the ramified slope $1/4$ require Borel–Laplace multisummation that we did not complete on the host; we report the layout and the determinant constraint and do not substitute a weaker proxy for the multipliers [CC4-1-STOKES]. This is the one computational leg of the program left open here (§11).

8.3. The integer-relation null as exclusion bounds. We tested the falsification target “the connection coefficient is an elementary Γ -quotient, a low-height constant combination, or algebraic”. The targets C_{EBR} and A are known to 169 digits; we ran a PSLQ battery at working precision 169 with acceptance tolerance 10^{-150} , leaving roughly 19 guard digits. The battery’s validity is demonstrated by three positive controls, all detected with the expected integer vectors: the Γ -reflection $\Gamma(1/3)\Gamma(2/3) = 2\pi/\sqrt{3}$, the algebraic relation $x^2 - 2 = 0$ for $\sqrt{2}$, and the linear relation $2\pi + 3 \log 2$.

For a basis of n terms, a spurious relation of integer height H produces a residual of order $H^{-(n-1)}$; for the search to be limited by the searched height rather than by precision we require $H^{-(n-1)}$ to remain above the tolerance, i.e. the precision-detection ceiling is $H_{\text{max}} \approx 10^{150/(n-1)}$. In every tier below the searched height lies well within this ceiling, so each null is a genuine exclusion bound: *any* integer relation must have height exceeding the searched H .

Tier	basis (n terms)	searched H	ceiling $10^{150/(n-1)}$	verdict
Γ -monomial	reflection-normalized Γ at $(1/24)\mathbb{Z}$, indep. ($n=8$)	10^9	$10^{21.4}$	null
Γ -monomial	overcomplete 11- Γ basis ($n=15$)	10^9	$10^{10.7}$	inter- Γ only
constants	$\{1, \pi, \log 2, \log 3, \gamma_E, \text{Cat}, \zeta_3, \zeta_5\}$ ($n=9$)	10^{12}	$10^{18.8}$	null
algebraic	minimal polynomial $\deg \leq 8$ ($n \leq 9$)	10^6	$10^{18.8}$	null

In the reflection-normalized independent Γ -basis there is no relation at all; in the overcomplete 11- Γ basis PSLQ recovers only genuine Gauss multiplication/reflection identities among the Γ -values, each with *coefficient zero on the target* (residuals $\sim 10^{-170}$) — an internal control showing the battery sees real Γ -structure without entangling the target. To 169 digits, then, C_{EBR} and A are not elementary Γ -quotients [EBR3-B-GAMMA], not

integer-linear combinations of $\{1, \pi, \log 2, \log 3, \gamma_E, \text{Catalan}, \zeta(3), \zeta(5)\}$ [EBR3-B-CONST], and not algebraic of degree ≤ 8 and height $\leq 10^6$ [EBR3-B-ALG]. This strengthens the ≥ 33 -digit Γ -quotient null of EBR-II to 169 digits. It is evidence of non-elementarity; it is not a proof of transcendence, and we do not present it as one.

9. LEAN CORES AND THE AXIOM AUDIT

The finitary, arithmetic sub-claims that admit machine-checked proofs are formalized in Lean 4 with Mathlib (`leanprover/lean4` v4.30.0; Mathlib rev `c5ea0035`). Four cores are proved: the degree-bound arithmetic $B_2 = 1 + 2 + 0 = 3$, $B_1 = 3 + 2 + 2 = 7$, both ≤ 20 ([CC4-LEAN-BOUNDS]); the pullback exponent integrality, that $s = t^2$ doubles $\{0, \frac{1}{2}, 1, \frac{3}{2}\}$ to the integers $\{0, 1, 2, 3\}$ ([CC4-LEAN-PULLBACK]); the eigenvalue-parity bookkeeping, that the R -spectrum is not negation-closed while the 0-spectrum is, encoded by angles in $\mathbb{Z}/6$ with negation $k \mapsto k + 3$ ([CC4-LEAN-PARITY]); and the parity core, that the 4-cycle `finRotate 4` is odd and so lies outside the alternating group, the group-theoretic heart of the 4-cycle closure ([CC4-LEAN-A1B]). A `#print axioms` audit confirms that every declaration depends only on a cone contained in `{propext, Classical.choice, Quot.sound}`, with no `sorryAx`; three of the bound lemmas depend on no axioms at all. These are the only PROVEN results in the paper.

10. FOUR-CLASS GRADE TABLE

Result	Grade	Evidence
L_2 explicit, singular locus $\{0, R\}$	V	[CC1-OP-L2]
Riemann data (3)	V	[CC1-RIEMANN]
∞ irregular, slope $1/4$	V	[CC1-INF-IRREGULAR]
Not a G -function (normalization-scoped)	V	[CC2-0-GFUNC], [CC2-0-QINT]
p -curvature non-nilpotent	V	[CC2-0-PCURV]
Trichotomy excludes G -function	S	[CC2-0-TRICHOTOMY]
Irreducible/minimal (two routes)	S	[CC1-IRREDUCIBLE], [CC2-0-FACTOR]
M_R semisimple pseudo-reflection	V	[CC2-2D-JORDAN], [CC4-1-NOLOG]
No invariant form ($\not\leq \mathrm{SO}_4, \mathrm{Sp}_4$)	V/S	[CC2-2A-SELFDUAL], [CC2-2B-SP4], [CC2-2C-SO4]
Tensor classes excluded	S	[CC2-2D-TENSOR]
$\eta = \sqrt{s}$ unique; 4-cycle closure	S	[CC4-A1-ETA-UNIQUE], [CC4-A1B-MONOMIAL-CLOSURE]
Primitive (two agreeing routes)	S	[CC4-0-ROUTE1-PRIMITIVE], [CC4-0-ROUTE2-PRIMITIVE]
Degree bounds $B_2=3, B_1=7 \leq 20$	S	[CC4-0B-BOUNDS]
Main: $G_{\mathrm{Gal}}(L_2)^\circ = \mathrm{SL}_4$	S	[CC4-0-SL4-VERDICT]
No Liouvillian solutions	V/S	[CC4-2-BRIDGE]
Connection coefficient to 169 digits	V	[CC4-1-C-120D]
Γ -quotient / constant / algebraic nulls	V	[EBR3-B-GAMMA], [EBR3-B-CONST], [EBR3-B-ALG]
Lean cores (4)	P	[CC4-LEAN-BOUNDS], [CC4-LEAN-PULLBACK], [CC4-LEAN-PARITY], [CC4-LEAN-A1B]
Transcendence of the connection coefficient	C	[CC1-DISCIPLINE]

11. OPEN PROBLEMS

We list the open directions, each tagged with the program op-code under which it is being pursued.

- (1) *Transcendence of the connection coefficient* (op:cc-3). The sole route is via a period-theoretic analysis, argued from first principles since the G -function clause is retired (§3). The 169-digit null of §8.3 is the current evidence of non-elementarity; the 169-digit value, the exact monodromy, and the Stokes layout are the inputs.

- (2) *Numerical Stokes multipliers* at the ramified slope $1/4$, via Borel–Laplace multisummation (§8.2).
- (3) *General d* . Only $d = 2$ is settled here; the irreducibility and the group computation at $d \geq 3$ require the order- $2d$ analogues of every step. Three ingredients are specific to $d = 2$ and do not transfer for free: the determining-factor orbit at infinity is a single 4-cycle (a transitive $\mathbb{Z}/2d$ -orbit at general d , so the slope-additivity factor enumeration grows), the target group becomes SL_{2d} with its own and larger maximal-subgroup list, and the *a priori* degree bounds of §6.5 must be recomputed from the order- $2d$ exponent data. The arithmetic conclusions are expected to persist but are not established here.
- (4) *The rigidity index* with the irregular point, completing the accessory count beyond the robust positivity used in §2.
- (5) *The rigidity dividing-line conjecture*. The precise statement — that the connection coefficient is elementary exactly on the rigid ($d = 1$) or resonant/reducible members and a genuine transcendental period generically for $d \geq 2$ — remains conjectural; the present work supplies the group side at $d = 2$ but not the arithmetic side.

REPRODUCIBILITY

All computations are reproducible from the accompanying package: the symbolic operator and singular-structure scripts, the p -curvature and factor-exclusion scripts, the twist and invariant-form searches, the primitivity routes and degree bounds, the 169-digit connection computation, the Stokes layout, and the integer-relation battery, each with a canonical content hash recorded in full in `HASH.MANIFEST.json`; together with the Lean project (buildable, with the axiom-audit file) and a snapshot of the claims ledger. A `README` gives a one-command verification path per artifact, and `verify.py` re-derives every embedded self-hash and file hash against the manifest (exit code 0).

The load-bearing frozen inputs are pinned here by SHA-256 prefix (full 64-hex digests in `HASH.MANIFEST.json`). The 169-digit integer-relation null `ebr3.b.pslq_results.json` begins `9a3f942def64737d...`; the connection coefficient `cc4_1_connection_results.json` begins `f3400831cc964464...`; and the claims-ledger snapshot begins `9e6f3fa5a2986cce...` (The ledger hash is non-circular: no claim references the bytes of this document.)

REFERENCES

- [1] Papanokechi, *An Edge–Borel Radius Theorem for Positive Polynomial Continued Fractions*, Zenodo, 10.5281/zenodo.20564079 (v1.0: 20564080).
- [2] Papanokechi, *The EBR Amplitude as a Connection Coefficient: Characterization and a Rigidity Dividing-Line Conjecture*, Zenodo, 10.5281/zenodo.20566465 (v1.2: 20571232).
- [3] M. van der Put and M. F. Singer, *Galois Theory of Linear Differential Equations*, Grundlehren der mathematischen Wissenschaften **328**, Springer, 2003.
- [4] Y. André, *G-functions and Geometry*, Aspects of Mathematics **E13**, Vieweg, 1989.

- [5] N. M. Katz, *Nilpotent connections and the monodromy theorem: applications of a result of Turrittin*, Publ. Math. IHÉS **39** (1970), 175–232.
- [6] J. J. Kovacic, *An algorithm for solving second order linear homogeneous differential equations*, J. Symbolic Comput. **2** (1986), 3–43.
- [7] H. R. P. Ferguson, D. H. Bailey, and S. Arno, *Analysis of PSLQ, an integer relation finding algorithm*, Math. Comp. **68** (1999), 351–369.
- [8] The Mathlib Community, *The Lean Mathematical Library*, CPP 2020, 367–381.